MF-CLIP: Leveraging CLIP as Surrogate Models for No-Box Adversarial Attacks

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***Abstract*—The vulnerability of Deep Neural Networks (DNNs) to adversarial attacks poses a significant challenge to their deployment in safety-critical applications. While extensive research has addressed various attack scenarios, the no-box attack setting—where adversaries have no prior knowledge, including access to training data of the target model—remains relatively underexplored despite its practical relevance. This work presents a systematic investigation into leveraging large-scale Vision-Language Models (VLMs), particularly CLIP, as surrogate models for executing no-box attacks. Our theoretical and empirical analyses reveal a key limitation in the execution of no-box attacks stemming from insufficient discriminative capabilities for direct application of vanilla CLIP as a surrogate model. To address this limitation, we propose MF-CLIP (Margin-based Fine-tuned CLIP), a novel framework that enhances CLIP's effectiveness as a surrogate model through margin-aware feature space optimization. Comprehensive evaluations across diverse architectures and datasets demonstrate that MF-CLIP substantially advances the state-of-the-art in no-box attacks, surpassing existing baselines by 15.23% on standard models and achieving a 9.52% improvement on adversarially trained models. Our code is made publicly available to facilitate reproducibility and future research in this direction.**

***Index Terms*—Adversarial attacks, vision-language models.**

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I. Introduction

Deep Neural Networks (DNNs) have demonstrated exceptional performance across various applications, showcasing impressive generalization capabilities. However, this success also raises significant concerns regarding the trustworthiness of AI systems [1], [2], [3], [4]. In particular, DNNs remain vulnerable to adversarial attacks, which involve subtle, often imperceptible perturbations to input images that exploit weaknesses in the models' decision boundaries [5], [6], [7], [8], [9], [10]. These attacks involve subtle, often imperceptible perturbations to input images that exploit inherent weaknesses in DNNs' decision boundaries across multiple attack scenarios: white-box, black-box, and no-box. Each scenario introduces unique challenges based on the attacker's level of knowledge about the target model, forming a continuum of decreasing information accessibility.

Among these, the no-box scenario, which lies at the extreme end of the knowledge spectrum, presents a particularly challenging threat model. In this setting, the attacker has zero knowledge about the target model—lacking access to its architecture, parameters, training methodology, and crucially, training data [11]. This distinguishes no-box attacks from the more commonly studied black-box attacks, where adversaries typically have at least partial knowledge about training data or can query the model for feedback. The no-box scenario represents a more realistic security concern in production environments where models are completely inaccessible behind APIs or embedded systems, offering a practical reflection of real-world security challenges in deployed machine learning systems.

Despite its practical significance, executing effective no-box attacks remains challenging due to the fundamental issue of transferability—the ability of adversarial examples crafted using one model to successfully deceive another. This transferability becomes especially difficult when operating with minimal knowledge about the target model. The effectiveness of no-box attacks hinges critically on the choice and configuration of surrogate models, which serve as proxies for generating adversarial examples in the absence of direct access to target models. However, this crucial aspect of surrogate model selection has been largely overlooked in existing literature, with most research focusing instead on sophisticated attack methodologies rather than the foundation upon which these attacks are built.

To address this gap, we establish comprehensive criteria for selecting and optimizing surrogate models specifically tailored for no-box adversarial attacks. Our approach reconceptualizes adversarial attacks as an image generation problem, specifically aiming to generate non-robust features that can effectively transfer across model boundaries. Drawing from established principles in image generation [12], we formulate a core principle: the ideal surrogate model should be trained on a diverse and comprehensive dataset that approximates the knowledge spectrum of potential target models, allowing it to generate perturbations that exploit common vulnerabilities across various architectures.

In this context, large-scale Vision-Language Models (VLMs) emerge as promising candidates, with CLIP [13] standing out for its exceptional versatility and rich representational capacity.[1] CLIP's training on massive and diverse image-text pairs positions it as an ideal surrogate model candidate for transferable attacks. However, our initial experiments reveal a counterintuitive finding: the direct application of vanilla CLIP as a surrogate model fails to yield improved performance, challenging the intuitive expectation. Through rigorous analysis, we discover that while CLIP possesses remarkable ability to represent diverse visual features across domains, it lacks the necessary discriminative capability essential for effective no-box adversarial attacks—a finding that reveals a fundamental limitation in using foundation models as-is for adversarial purposes.

To further investigate this phenomenon, we conduct a theoretical analysis examining the relationship between discriminative capacity and the inter-class gap, quantifiable by a metric we term the margin. This analysis, complemented by visual examination of CLIP's feature space distribution, confirms that while CLIP excels in representational capacity, it exhibits significant limitations in discriminative power. Specifically, CLIP's feature space reveals compressed inter-class distances within specific domains, creating a fundamental barrier to generating effective domain-specific adversarial perturbations.

Building on these insights, we propose a novel fine-tuning methodology named **Margin-based Fine-tuned CLIP (MF-CLIP)**. This approach is specifically designed to enhance CLIP's discriminative abilities through targeted margin optimization, transforming it into an efficient surrogate model for no-box adversarial attack scenarios. Crucially, the images used in this fine-tuning process—which we refer to as target images—have no overlap with the training sets of the corresponding target models, ensuring strict compliance with the assumptions of the no-box scenario.

Our comprehensive experimental evaluation across diverse datasets and target models demonstrates that MF-CLIP substantially advances the state-of-the-art in no-box adversarial attacks, outperforming the best baseline method by an average of **15.23%** on standard models (Fig. 1) and achieving a **9.52%** improvement on adversarially trained models. These results not only validate our approach but also highlight the critical,

[1]For simplicity, "CLIP" refers to CLIP's image encoder unless otherwise noted throughout this paper.

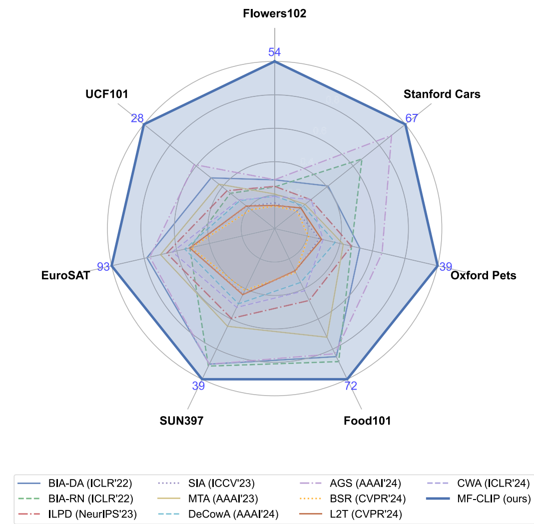


Fig. 1. Comparing MF-CLIP's attack success rate (ASR) against state-of-the-art methods across seven datasets. Results represent average performance across three target models (EfficientNet-B0, RegNetX-1.6GF, and ResNet-18). The visualization demonstrates MF-CLIP's consistent and substantial performance advantages across all datasets.

yet often overlooked, role of surrogate model optimization in adversarial machine learning.

Our main contributions are summarized as follows:

- We provide a systematic investigation into the often-overlooked role of surrogate models in no-box adversarial attacks, establishing theoretical and empirical foundations for leveraging large-scale VLMs with diverse data exposure to improve attack efficacy in challenging scenarios.
- We propose MF-CLIP, a margin-based fine-tuned version of CLIP that addresses the fundamental limitation of compressed inter-class distances in foundation models, thereby enhancing discriminative power for domain-specific adversarial attacks while preserving CLIP's rich representational capabilities.
- We demonstrate through extensive experimentation the superiority of MF-CLIP over existing attack methods across a range of models, datasets, and defensive strategies, offering novel insights into the utility of surrogate model fine-tuning in adversarial settings and establishing new benchmarks for no-box attacks.

II. Background and Related Work

A. Vision-Language Models

Recent advances in Natural Language Processing (NLP) have led to the creation of large-scale models that enjoy a broad spectrum of knowledge [14]. In computer vision (CV), the focus has shifted toward developing multimodal large-scale models that integrate visual and textual information. While some models, such as Qwen 2.5 [15] and LLaVA 1.5 [16], extend NLP-centric models to support multimodal functionalities, they remain predominantly optimized for text. In contrast, CLIP [13] achieves a more balanced integration of visual and textual modalities, as demonstrated by the widespread standalone use of its image encoder in various CV tasks. This highlights CLIP's effectiveness as a bridge between language and vision. Recent work has also explored the adversarial robustness properties of VLMs, including investigations into their transferability characteristics across different model architectures [17]. These studies provide valuable insights into the broader security implications of vision-language models in adversarial settings.

B. Adversarial Attacks: Evolution and Classification

Adversarial attacks, first identified by Szegedy et al. [5], exploit vulnerabilities in neural networks by introducing carefully crafted perturbations to input data that cause misclassification while remaining imperceptible to human observers. These attacks can be broadly categorized based on implementation strategy (query-based versus transfer-based) and knowledge assumptions (white-box, black-box, and no-box).

Query-based attacks rely on iterative feedback from the target model to refine adversarial examples, requiring direct access to model outputs. In contrast, transfer-based attacks—the focus of this paper—generate adversarial examples using a surrogate model and exploit the transferability property to attack target models without requiring query access. This approach is particularly relevant in real-world scenarios where direct interaction with target models may be restricted or monitored.

The evolution of transfer-based attacks can be delineated into three distinct phases, representing a continuum of decreasing knowledge about the target model:

- **White-box Setting**: The adversary has complete access to the target model's architecture, parameters, and training data. This represents the strongest attack scenario, where adversarial examples can be optimized directly against the target model [18], [19].
- **Black-box Setting**: The adversary has no access to the model's architecture or parameters but retains knowledge about or access to the training data distribution. This scenario has received extensive research attention, with numerous techniques developed to enhance transferability across architectures [20], [21].
- **No-box Setting**: Introduced by Li et al. [11], this represents the most challenging scenario where the adversary lacks access to the model's architecture, parameters, and crucially, its training data. This setting models realistic security threats against deployed systems where attackers have minimal information about the target model.

The progression through these phases reflects increasing practical relevance but also growing technical difficulty, as each reduction in knowledge significantly constrains the attacker's ability to craft effective adversarial examples.

C. Adversarial Attack Methodologies

Recent advancements in adversarial attacks have explored various settings, from white-box to no-box scenarios, with each introducing unique challenges and requiring innovative solutions [18], [19], [20], [22], [23]. To address these challenges, several distinct approaches have emerged in the literature:

- **Gradient Enhancement Methods**: Techniques like MI-FGSM [20] enhance gradients to improve adversarial effectiveness.
- **Input Augmentation Strategies**: Methods such as SIA [24], DeCowA [25], L2T [22], and BSR [23] apply input transformations to improve attack generalization across models.
- **Intermediate Layer Attacks**: These approaches target intermediate layers in surrogate models, often achieving superior transferability compared to attacks on the final layer. Notable methods include ILA [26], FIA [27], and ILPD [28].
- **Generator-based Approaches**: Generator models, such as BIA [29], are used to produce adversarial examples directly.
- **Layer Modification Techniques**: Strategies like AGS [30] modify specific layers within surrogate models to boost attack effectiveness.

These categories represent a spectrum of innovative approaches that address the evolving challenges of adversarial attacks in various access settings. Our proposed method emphasizes fine-tuning CLIP and employs a generator model to generate adversarial images.

III. Proposed Method

In this section, we first establish the theoretical foundations of our approach by outlining the threat model and problem formulation. We then provide a detailed analysis of CLIP's capabilities and limitations in the context of adversarial attacks, culminating in the introduction of our proposed method, MF-CLIP.

A. Preliminaries

a) Threat Model: Our work addresses a practical no-box attack setting that realistically models security threats against deployed machine learning systems. In this setting, the adversary has access to a surrogate model but possesses no information about the target model's architecture, parameters, or—crucially—its training data distribution. This represents a significant departure from conventional black-box settings, where knowledge of or access to the target's training distribution is typically assumed.

The no-box setting creates a substantial knowledge gap between surrogate and target models, making the transferability of adversarial examples particularly challenging. The attacker must indirectly compromise the target using only the surrogate model as a proxy, though they can leverage additional data (explicitly guaranteed not to overlap with the target model's training set) to enhance their attack strategy. This model represents real-world scenarios where attackers might target commercially deployed systems without access to proprietary training data or model details.

b) Formulation: We formally define the no-box attack scenario as follows: Given a set of original images x with corresponding ground truth labels y, we aim to craft adversarial examples x' using a pre-trained surrogate model f. The objective is to generate x' that are visually indistinguishable from x (preserving semantic content) while inducing misclassification in the target model g, such that $g(x') \neq y$.

We focus on untargeted attacks, where the goal is to cause any misclassification rather than forcing a specific target class. Additionally, we constrain the adversarial perturbations $\delta = x' - x$ to satisfy $\|\delta\|_\infty \leq \epsilon$, where ϵ represents a pre-defined bound on the perturbation magnitude, ensuring that the adversarial examples remain visually similar to the original images.

For the surrogate model f with parameters θ_f, the objective is to maximize its loss function $J(x', y)$ with respect to the adversarial examples x':

$$\arg\max_{x'} J(x', y; \theta_f), \quad \text{s.t.} \quad \|x' - x\|_\infty \leq \epsilon. \tag{1}$$

To efficiently generate these adversarial examples, we employ a generator network G trained to produce perturbations that maximize the surrogate model's loss. The adversarial examples are then obtained as $x' = G(x)$, with the generator implicitly enforcing the perturbation constraint during training.

B. Analysis of CLIP's Representational and Discriminative Capabilities

Before introducing our proposed method, we first analyze CLIP's fundamental properties to understand its potential and limitations as a surrogate model for adversarial attacks. This analysis provides the theoretical foundation for our approach.

1) Overview of CLIP Architecture and Training Methodology: CLIP (Contrastive Language-Image Pre-training) comprises two parallel encoders: an image encoder f_ψ and a text encoder f_ϕ. The image encoder f_ψ maps input images x to dense embeddings $f_\psi(x) \in \mathbb{R}^d$, while the text encoder f_ϕ projects textual descriptions y to the same embedding space $f_\phi(y) \in \mathbb{R}^d$.

CLIP employs a similarity function $h(x, y) = \text{sim}(f_\psi(x), f_\phi(y))$ to measure alignment between visual and textual representations. This similarity is defined as the normalized dot product (cosine similarity) of the respective embeddings:

$$h(x, y) = \frac{\langle f_\psi(x), f_\phi(y)\rangle}{\|f_\psi(x)\|\|f_\phi(y)\|}. \tag{2}$$

During training, CLIP optimizes a contrastive loss function with a temperature parameter τ across mini-batches of size N. This loss function encourages matching image-text pairs to have high similarity while minimizing similarity between non-matching pairs:

$$\begin{aligned} L_S = &\frac{1}{N}\sum_{i=1}^{N}\left[-\log\left(\frac{\exp(h(x_i, y_i)/\tau)}{\sum_{j=1}^{N}\exp(h(x_j, y_i)/\tau)}\right)\right] \\ &+ \frac{1}{N}\sum_{i=1}^{N}\left[-\log\left(\frac{\exp(h(x_i, y_i)/\tau)}{\sum_{j=1}^{N}\exp(h(x_i, y_j)/\tau)}\right)\right]. \end{aligned} \tag{3}$$

This can be reformulated to highlight the relative differences in similarities:

$$\begin{aligned} L_S = &\frac{1}{N}\sum_{i=1}^{N}\log\left(\sum_{j=1}^{N}\exp\left(\frac{h(x_j, y_i) - h(x_i, y_i)}{\tau}\right)\right) \\ &+ \frac{1}{N}\sum_{i=1}^{N}\log\left(\sum_{j=1}^{N}\exp\left(\frac{h(x_i, y_j) - h(x_i, y_i)}{\tau}\right)\right). \end{aligned} \tag{4}$$

2) Theoretical Analysis of Margin in CLIP's Feature Space: The inter-class margin—the separation between different classes in feature space—plays a critical role in determining a model's discriminative power and, consequently, its effectiveness in classification tasks. In the context of adversarial attacks, models with smaller margins between classes are generally more vulnerable, as smaller perturbations can cross decision boundaries.

For machine learning models, the margin quantifies the inherent separation between distinct categories within the feature space [31]. It represents the difference in similarity between positive pairs (samples from the same class) and negative pairs (samples from different classes), typically expressed as $h(x, x^+) - h(x, x^-)$, where h denotes the similarity function.

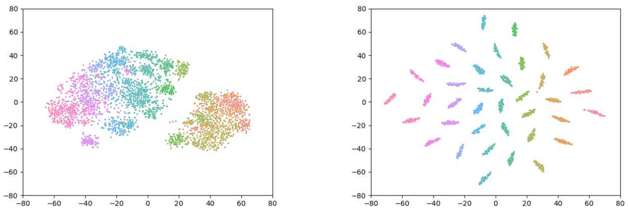
In CLIP's feature space, for an image x with true label y and a different label y', the margin Δ can be mathematically formulated as:

$$\begin{aligned} \Delta &= h(x, y) - h(x, y') \\ &= \frac{1}{2}\left(2 - \|f_\psi(x) - f_\phi(y)\|^2\right) \\ &\quad - \frac{1}{2}\left(2 - \|f_\psi(x) - f_\phi(y')\|^2\right) \\ &= \frac{1}{2}\left(\|f_\psi(x) - f_\phi(y')\|^2 - \|f_\psi(x) - f_\phi(y)\|^2\right). \end{aligned} \tag{5}$$

This formulation reveals that CLIP's contrastive loss implicitly optimizes the margin between classes. By rewriting Equation (4) in terms of Euclidean distances in the feature space:

$$\begin{aligned} &L_S \\ &= \frac{1}{2N}\sum_{i=1}^{N} \\ &\quad \log\left(\sum_{j=1}^{N}\exp\left(\frac{\|f_\psi(x_i) - f_\phi(y_i)\|^2 - \|f_\psi(x_j) - f_\phi(y_i)\|^2}{\tau}\right)\right) \\ &\quad + \frac{1}{2N}\sum_{i=1}^{N} \\ &\quad \log\left(\sum_{j=1}^{N}\exp\left(\frac{\|f_\psi(x_i) - f_\phi(y_i)\|^2 - \|f_\psi(x_i) - f_\phi(y_j)\|^2}{\tau}\right)\right). \end{aligned} \tag{6}$$

From this reformulation, it becomes evident that CLIP's training objective intrinsically aims to maximize the margin between classes in its feature space. However, our analysis reveals a critical limitation: while CLIP optimizes margins across its entire training distribution of 400 million image-text pairs, the resulting margins for specific downstream domains may be suboptimal due to the generalist nature of its training.



(a) Vanilla CLIP (b) MF-CLIP

Fig. 2. The t-SNE visualization results for the (a) vanilla CLIP and (b) MF-CLIP on the 37-class OxfordPet dataset.

3) Empirical Analysis through Feature Space Visualization: To complement our theoretical analysis with empirical evidence, we visualize CLIP's feature space using t-SNE [32], as illustrated in Fig. 2. Focusing on the 37-class OxfordPet dataset, our visualization reveals two key insights:

a) Strong Representational Capacity: : CLIP's embeddings naturally organize into two distinct high-level clusters corresponding to cats and dogs, demonstrating the model's ability to capture broad semantic relationships without fine-tuning. This confirms CLIP's exceptional representational capacity, which allows it to effectively model diverse visual concepts.

b) Limited Discriminative Power: : Despite successfully separating high-level categories, CLIP exhibits narrow margins between neighboring fine-grained classes (specific breeds) within each cluster. This compressed inter-class spacing reveals a critical limitation in CLIP's discriminative capabilities for fine-grained classification tasks.

This visualization confirms our theoretical hypothesis: while CLIP excels at representing broad semantic relationships due to its diverse training, it lacks the fine-grained discriminative capabilities necessary for specific downstream tasks. In the context of adversarial attacks, this limitation is particularly problematic, as effective attacks require exploiting precise decision boundaries between classes—boundaries that are poorly defined when inter-class margins are compressed.

The insufficient discriminative power within specific domains represents a fundamental barrier to CLIP's effectiveness as a surrogate model for no-box adversarial attacks. This finding motivates our proposed approach: enhancing CLIP's discriminative capabilities through targeted fine-tuning while preserving its rich representational power.

4) Theoretical Mechanism of Enhanced Cross-Model Attack Transferability: Our comprehensive analysis establishes the fundamental theoretical mechanism underlying MF-CLIP's superior cross-model attack transferability, addressing a critical gap in understanding how surrogate model optimization enhances adversarial transferability across diverse architectures.

a) Margin-Transferability Relationship: The core principle lies in the relationship between inter-class margins and adversarial transferability. In adversarial machine learning, the vulnerability of neural networks stems from the existence of adversarial subspaces—regions in the input space where small perturbations can cause misclassification [5]. These subspaces are fundamentally determined by the geometry of decision boundaries in feature space. When a surrogate model has larger inter-class margins, it naturally identifies more robust and generalizable adversarial directions that exploit common geometric vulnerabilities shared across different model architectures.

b) Cross-Architecture Generalization: By explicitly enlarging inter-class margins through margin-based fine-tuning, MF-CLIP creates more pronounced and stable decision boundaries that generalize better across different model architectures. Mathematically, this can be understood through the principle that larger margins in the feature space correspond to simpler and more robust decision functions with superior generalization properties [31]. When applied to adversarial transferability, this means that adversarial perturbations crafted using a high-margin surrogate model are more likely to exploit fundamental, architecture-independent vulnerabilities rather than model-specific artifacts.

c) Robust Perturbation Generation: Larger margins in the surrogate model's feature space lead to more robust adversarial perturbations that can successfully exploit similar vulnerabilities in diverse target models. This occurs because the margin-enhanced feature space better captures the underlying data manifold structure, enabling the generation of perturbations that move samples across decision boundaries in ways that are consistent with the fundamental geometric properties of the problem domain, thereby fundamentally enhancing cross-model attack transferability.

C. MF-CLIP: Enhancing CLIP's Discriminative Capabilities

Building on our theoretical and empirical analysis, we propose Margin-based Fine-tuned CLIP (**MF-CLIP**), a novel approach designed to enhance CLIP's effectiveness as a surrogate model for no-box adversarial attacks by specifically addressing its limited discriminative capabilities.

The key insight driving our approach is that while CLIP possesses exceptional representational capacity due to its training on diverse data, its discriminative power within specific domains—crucial for generating effective adversarial examples—requires enhancement. MF-CLIP achieves this through a targeted fine-tuning process that explicitly optimizes inter-class margins while preserving CLIP's rich representational capabilities.

Fig. 3 illustrates our proposed two-stage pipeline for MF-CLIP. In the first stage, we enhance CLIP's discriminative capabilities through margin-aware fine-tuning. In the second stage, we leverage the enhanced model as a surrogate to generate transferable adversarial examples.

1) Stage 1: Margin-Aware Fine-Tuning: In the first stage, we utilize CLIP as a feature extractor f_θ with parameters θ to extract d-dimensional embeddings from input images x. To enhance its discriminative capabilities, we augment CLIP with a supervisory head h, implemented as a fully connected network, and fine-tune the entire model using a margin-based loss function.

Specifically, we employ the Additive Angular Margin Loss [31], a state-of-the-art approach for enhancing inter-class

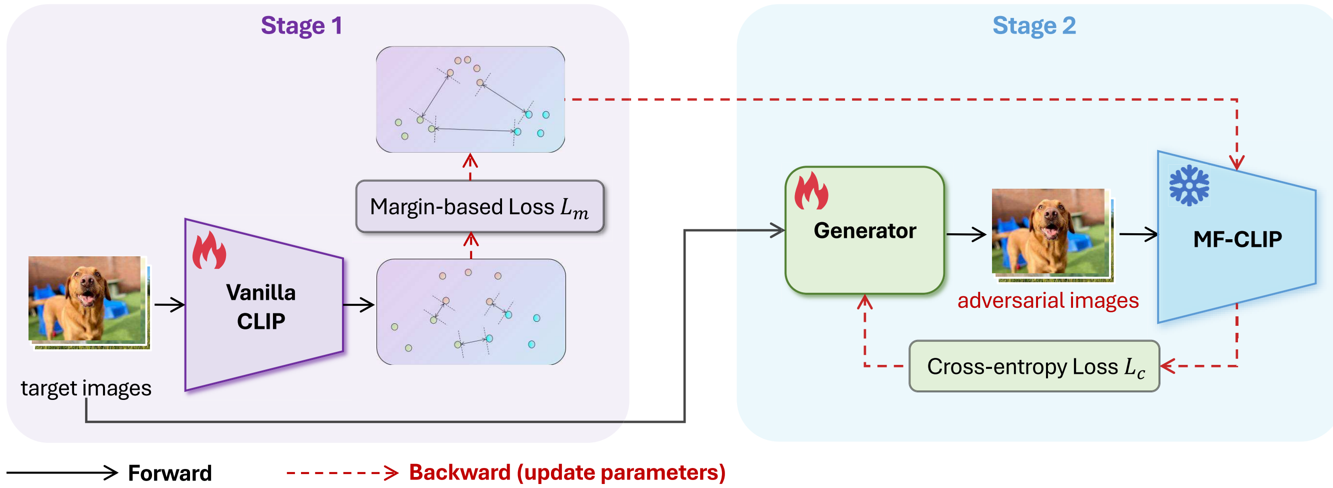


Fig. 3. The proposed MF-CLIP framework. Our approach consists of two main stages: (1) Margin-aware fine-tuning, which enhances CLIP's discriminative capabilities by optimizing inter-class margins while preserving its representational power; and (2) Adversarial example generation, where the fine-tuned model serves as a surrogate to generate highly transferable adversarial examples. In the diagram, black solid lines represent the forward process, while red dashed lines indicate the backward process (parameter updates).

separability in embedding spaces:

$$L_m = -\log \frac{e^{f_\theta(x_i)(\cos(\alpha_{y_i}+m))}}{e^{f_\theta(x_i)(\cos(\alpha_{y_i}+m))} + \sum_{j=1, j\neq y_i}^{n} e^{f_\theta(x_i)\cos\alpha_j}}, \quad (7)$$

where h is parameterized by weights W and a bias b (set to 0). Here, W_j denotes the j-th column of W, $f_\theta(x_i)$ represents the embedding of the i-th image from the y_i-th class, and n indicates the total number of classes in the target dataset.

With $b = 0$, the logit can be expressed as $W_j^T f_\theta(x_i) = \|W_j\|\|f_\theta(x_i)\| \cos\alpha_j$, where α_j represents the angle between weight vector W_j and feature vector $f_\theta(x_i)$. The margin penalty m explicitly increases the angular separation between classes, enhancing the model's discriminative power.

Crucially, to adhere to the constraints of the no-box setting, we fine-tune using target images x that are specifically selected for the attack and guaranteed to have no overlap with the training sets of the target models. This ensures that our approach remains applicable in realistic scenarios where attackers have no access to the target model's training data.

2) Stage 2: Adversarial Example Generation: In the second stage, we employ the fine-tuned CLIP model $f_{\hat{\theta}}$ as a surrogate for generating adversarial examples. Rather than directly optimizing perturbations through iterative methods, we train a generator network G to produce adversarial examples efficiently.

The generator G is based on a U-Net architecture, incorporating downsampling and upsampling layers with skip connections and ResBlocks. This design allows the generator to learn complex perturbation patterns while maintaining the spatial structure of the input images. The generator is trained to induce misclassification in the fine-tuned surrogate model:

$$L_c = -\mathcal{C}(F(G(x)), y), \quad \text{s.t.} \quad \|G(x) - x\|_\infty \leq \epsilon, \quad (8)$$

where $\mathcal{C}$ denotes the cross-entropy loss, and $F = f_{\hat{\theta}} \circ h$ represents the composite network of the fine-tuned feature extractor and supervisory head. The constraint $\|G(x) - x\|_\infty \leq \epsilon$ ensures that the perturbations remain bounded, preserving the visual similarity between original and adversarial images.

By combining margin-aware fine-tuning with efficient adversarial example generation, MF-CLIP addresses the fundamental limitations of using vanilla CLIP as a surrogate model for no-box attacks. The margin-based fine-tuning enhances CLIP's discriminative capabilities in the target domain, creating more pronounced decision boundaries that can be effectively exploited to generate transferable adversarial examples. Meanwhile, the generator network efficiently learns to produce structured perturbations that maximize the likelihood of successful transfer to unknown target models.

IV. Experiments

This section presents comprehensive experimental evaluations of our proposed approach. We first describe our experimental setup, then demonstrate the effectiveness of MF-CLIP through extensive comparisons with state-of-the-art methods on both standard and adversarially trained models, followed by detailed analyses.

A. Experimental Settings

1) Datasets and Models: We conduct experiments on seven diverse, high-resolution vision datasets: OxfordPets [33], Flowers102 [34], StanfordCars [35], Food101 [36], SUN397 [37], EuroSAT [38], and UCF101 [39]. For target models, we employ ResNet-18 [40], EfficientNet-B0 [41], and RegNetX-1.6GF [42].

2) Implementation Details: We fine-tune CLIP using SGD optimizer with a batch size of 128 and an initial learning rate of 0.01, following a cosine annealing schedule over 300 epochs. The generator is trained using AdamW optimizer with a batch size of 64 and an initial learning rate of 0.0001 for 90 epochs, also with cosine annealing. We set the margin parameter $m = 0.15$ and constrain the ℓ_∞-norm perturbation to $\epsilon = 16/255$. Test set images are used as target images

TABLE I

ASR PERFORMANCE COMPARISON OF OUR METHOD AGAINST STATE-OF-THE-ART APPROACHES ACROSS VARIOUS DATASETS AND TARGET MODELS. THE HIGHER VALUES INDICATE BETTER PERFORMANCE. THE ↑ ROW SHOWS THE IMPROVEMENT OF OUR METHOD OVER THE BEST BASELINE FOR EACH DATASET AND TARGET MODEL. AVERAGE INDICATES THE MEAN PERFORMANCE OF EACH METHOD ACROSS ALL DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Method	Flowers102	Stanford Cars	Oxford Pets	Food101	SUN397	EuroSAT	UCF101	*Average*
EfficientNet-B0	BIA-DA (ICLR 2022)	15.87	19.76	23.88	60.62	37.90	70.20	19.30	35.36
	BIA-RN (ICLR 2022)	13.60	55.95	19.46	63.28	38.18	49.75	11.76	36.00
	ILPD (NeurIPS 2023)	15.87	21.94	20.93	37.04	25.49	57.46	12.19	27.27
	SIA (ICCV 2023)	10.39	14.08	13.74	22.34	19.77	43.19	8.49	18.86
	MTA (AAAI 2023)	11.45	17.65	18.75	52.06	29.01	64.31	14.64	29.70
	DeCowA (AAAI 2024)	12.99	18.65	16.22	28.56	22.49	47.86	9.60	22.34
	AGS (AAAI 2024)	17.34	64.47	24.56	59.97	36.90	69.65	18.85	41.68
	BSR (CVPR 2024)	10.35	14.08	13.98	22.14	19.59	42.06	8.19	18.63
	L2T (CVPR 2024)	10.90	16.30	14.10	21.00	21.20	44.80	8.60	19.56
	CWA (ICLR 2024)	14.13	25.21	18.12	32.26	26.84	55.17	11.84	26.22
	MF-CLIP (ours)	**46.89**	**66.68**	**30.50**	**70.05**	**40.99**	**91.83**	**28.18**	**53.59**
	↑	+29.55	+2.21	+5.94	+6.77	+2.81	+21.63	+8.88	+11.91
RegNetX-1.6GF	BIA-DA (ICLR 2022)	25.46	42.18	22.95	64.93	37.24	82.79	13.53	41.30
	BIA-RN (ICLR 2022)	20.99	51.16	22.21	69.91	37.03	63.68	12.21	39.60
	ILPD (NeurIPS 2023)	17.30	23.37	19.84	38.62	27.12	72.83	11.55	30.09
	SIA (ICCV 2023)	10.80	14.53	10.60	23.94	20.84	63.69	6.66	21.58
	MTA (AAAI 2023)	15.23	18.89	17.93	57.45	25.48	72.32	10.84	31.16
	DeCowA (AAAI 2024)	12.42	16.83	15.21	29.09	23.20	64.10	8.67	24.22
	AGS (AAAI 2024)	15.59	64.41	26.96	59.97	36.33	74.67	14.91	41.83
	BSR (CVPR 2024)	10.43	14.07	9.57	23.38	20.48	62.14	6.32	20.91
	L2T (CVPR 2024)	9.20	17.00	12.50	25.00	19.60	59.60	4.80	21.10
	CWA (ICLR 2024)	14.82	24.38	14.20	34.95	27.77	71.32	9.70	28.16
	MF-CLIP (ours)	**61.39**	**75.60**	**43.91**	**74.67**	**38.89**	**92.86**	**28.71**	**59.43**
	↑	+35.93	+11.19	+16.95	+4.76	+1.65	+10.07	+13.80	+17.60
ResNet-18	BIA-DA (ICLR 2022)	6.01	19.76	13.79	57.05	31.32	65.11	8.30	28.76
	BIA-RN (ICLR 2022)	5.85	26.45	13.06	56.50	32.82	31.43	4.68	24.40
	ILPD (NeurIPS 2023)	7.88	9.97	13.90	26.83	17.94	53.95	7.01	19.64
	SIA (ICCV 2023)	3.82	5.48	7.44	15.04	10.81	39.85	2.91	12.19
	MTA (AAAI 2023)	6.94	10.12	12.18	45.43	22.26	58.27	10.52	23.67
	DeCowA (AAAI 2024)	6.09	8.16	11.69	19.11	13.37	41.52	4.31	14.89
	AGS (AAAI 2024)	14.54	49.93	24.94	58.50	33.25	68.80	18.19	38.31
	BSR (CVPR 2024)	4.10	5.75	6.62	14.90	10.46	38.17	2.88	11.84
	L2T (CVPR 2024)	4.20	8.40	8.80	14.20	13.10	40.70	4.90	13.47
	CWA (ICLR 2024)	6.09	10.24	10.19	22.46	16.29	50.06	5.47	17.26
	MF-CLIP (ours)	**56.07**	**55.69**	**41.67**	**70.14**	**38.44**	**91.77**	**27.68**	**54.49**
	↑	+41.53	+5.76	+16.73	+11.64	+5.62	+22.97	+9.49	+16.18

unless otherwise specified. All experiments were conducted on a server equipped with 8 NVIDIA A800 GPUs. We note that this is not the minimum requirement, and the proposed method can be effectively implemented on more accessible hardware, such as a single NVIDIA 3090 GPU, as detailed in our computational efficiency analysis (Section IV-D1).

a) Baselines and Evaluation Metric: We compare against state-of-the-art methods including AGS [30], BSR [23], CWA [43], ILPD [28], L2T [22], MTA [44], and SIA [24], implemented using TransferAttack.[2] We also include BIA [29] using official weights. All methods use ResNet-50 as the surrogate backbone, except CWA (ResNet-50/34 ensemble) and MTA (ResNet-MTA-18). Performance is measured using Attack Success Rate (ASR), defined as:

$$\mathrm{ASR} = \mathrm{Acc}_{\mathrm{clean}} - \mathrm{Acc}_{\mathrm{adv}} \tag{9}$$

where $\mathrm{Acc}_{\mathrm{clean}}$ is the accuracy on clean images and $\mathrm{Acc}_{\mathrm{adv}}$ is the accuracy on adversarial examples. Note that some studies define ASR as $1 - \mathrm{Acc}_{\mathrm{adv}}$, which can lead to higher reported values. We choose our definition to provide a direct measure of the attack's impact by explicitly showing the accuracy degradation.

[2]https://github.com/Trustworthy-AI-Group/TransferAttack

B. Evaluations on Normal Models

Table I presents comprehensive results across all datasets and target models. The row denoted by ↑ represents the improvement of our method over the best-performing baseline for each scenario. Notably, all values in this row are positive, indicating that MF-CLIP consistently outperforms existing methods across all test conditions. Quantitatively, MF-CLIP achieves average improvements of **11.91%**, **17.60%**, and **16.18%** over the best baseline for EfficientNet-B0, RegNetX-1.6GF, and ResNet-18, respectively. These substantial margins demonstrate the effectiveness of our approach in addressing the challenges of no-box attacks.

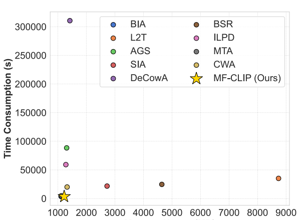
C. Evaluations on Adversarially Trained Models

We further evaluate all methods against models adversarially trained with PGD-10 [19] using an ϵ-radius. As shown in Table II, certain baseline methods yield negative ASR values, indicating that the adversarial accuracy surpasses the clean accuracy of the robust models. This outcome aligns with the known trade-off in adversarial training, where increased robustness often leads to a reduction in clean accuracy [45], highlighting the relative limitations of these attack methods. Notably, even with the enhanced robustness of these target

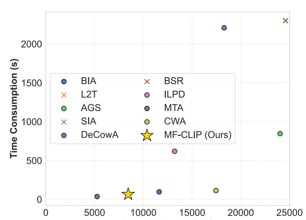
TABLE II

ASR Performance Comparison of Our Method Against State-of-the-Art Approaches on the PGD-10 Adversarially Trained Target Models. The Negative Values Indicate the Adversarial Accuracy Exceeds the Clean Accuracy of the Adversarially Trained Models. Average Indicates the Mean Performance of Each Method Across all Datasets. The Best Results are Highlighted in Bold

	Method	**Flowers102**	**Stanford Cars**	**Oxford Pets**	**Food101**	**SUN397**	**EuroSAT**	**UCF101**	***Average***
EfficientNet-B0	BIA-DA (ICLR 2022)	0.93	1.96	1.01	-9.99	-2.95	-24.94	0.13	-4.84
	BIA-RN (ICLR 2022)	1.02	1.75	0.74	-17.53	-3.41	-34.79	-0.11	-7.48
	ILPD (NeurIPS 2023)	0.41	1.08	0.76	-12.21	-3.17	-31.96	-0.19	-6.47
	SIA (ICCV 2023)	0.20	0.75	0.16	-10.74	-2.75	-35.74	-0.66	-6.97
	MTA (AAAI 2023)	0.41	1.24	0.71	-9.38	-1.46	-34.00	-0.11	-6.08
	DeCowA (AAAI 2024)	0.28	0.36	0.55	-10.13	-3.28	-36.81	-0.37	-7.06
	AGS (AAAI 2024)	2.11	5.78	2.26	-4.36	0.17	-7.86	1.06	-0.12
	BSR (CVPR 2024)	0.08	0.46	0.41	-10.62	-2.87	-36.36	-0.42	-7.05
	L2T (CVPR 2024)	1.05	0.37	0.42	-11.26	-1.10	-32.65	0.41	-6.10
	CWA (ICLR 2024)	0.37	0.56	0.30	-12.60	-3.36	37.72	0.26	3.32
	MF-CLIP (ours)	**4.50**	**12.26**	**2.32**	**1.13**	**2.19**	**55.85**	**16.39**	**13.52**
	↑	+2.39	+6.48	+0.06	+15.49	+2.02	+63.71	+15.33	+10.20
RegNetX-1.6GF	BIA-DA (ICLR 2022)	1.02	1.54	0.87	-2.74	1.23	9.31	0.00	1.60
	BIA-RN (ICLR 2022)	0.81	1.77	1.01	-2.97	2.45	-14.81	-0.11	-1.69
	ILPD (NeurIPS 2023)	-0.16	-0.24	0.71	-4.27	0.19	-6.43	-1.30	-1.64
	SIA (ICCV 2023)	-0.16	-0.24	0.71	-4.27	0.19	-6.43	-1.30	-1.64
	MTA (AAAI 2023)	0.41	-0.25	0.68	-3.55	-0.73	-1.05	-0.69	-0.74
	DeCowA (AAAI 2024)	-0.37	-0.22	0.57	-3.90	-0.70	-10.68	-0.95	-2.32
	AGS (AAAI 2024)	1.34	7.01	1.99	1.63	6.06	10.25	4.18	4.64
	BSR (CVPR 2024)	-0.45	-0.53	0.08	-4.68	-0.83	-12.40	-1.27	-2.87
	L2T (CVPR 2024)	0.52	-0.74	0.16	-4.48	0.07	-6.60	-1.00	-1.58
	CWA (ICLR 2024)	-0.49	-0.37	0.16	-4.65	-3.09	-5.98	1.42	-1.86
	MF-CLIP (ours)	**9.05**	**12.36**	**13.16**	**8.24**	**13.32**	**28.12**	**29.68**	**14.99**
	↑	+7.71	+5.35	+11.17	+6.61	+7.26	+17.87	+25.50	+10.35
ResNet-18	BIA-DA (ICLR 2022)	0.20	-0.42	0.74	4.93	-2.74	16.28	9.31	4.04
	BIA-RN (ICLR 2022)	0.20	-0.67	0.68	1.58	-2.97	-9.68	-3.09	-1.99
	ILPD (NeurIPS 2023)	0.12	-1.53	0.76	0.45	-1.71	-2.19	-2.14	-0.89
	SIA (ICCV 2023)	-0.53	-1.93	0.00	-1.11	-1.56	-6.10	-2.78	-2.00
	MTA (AAAI 2023)	-0.41	-1.89	0.16	-0.18	-2.80	3.70	-0.66	-0.30
	DeCowA (AAAI 2024)	-0.57	-1.85	0.57	-0.38	-1.87	-6.47	-2.62	-1.88
	AGS (AAAI 2024)	0.53	0.91	1.88	7.45	0.90	14.81	3.15	4.23
	BSR (CVPR 2024)	-0.49	-1.64	0.11	-1.29	-1.66	-6.70	-3.01	-2.10
	L2T (CVPR 2024)	-0.36	-2.13	0.62	-1.38	-0.37	-4.93	-2.89	-1.43
	CWA (ICLR 2024)	-0.20	-1.68	0.35	-0.40	-1.69	-6.84	3.09	-1.05
	MF-CLIP (ours)	**5.30**	**1.46**	**5.89**	**14.15**	**4.70**	**38.70**	**15.01**	**12.26**
	↑	+4.77	+0.55	+4.01	+6.70	+3.80	+23.89	+11.86	+8.03



(a) Batch size = 1



(b) Batch size = 128

Fig. 4. Computational efficiency comparison. Time consumption vs. memory usage at different batch sizes. The symbol indicates out of memory failures.

models, MF-CLIP achieves consistent positive gains across all scenarios, outperforming the strongest baseline methods by margins of **10.20%**, **10.35%**, and **8.03%** on EfficientNet-B0, RegNetX-1.6GF, and ResNet-18, respectively. These consistent gains demonstrate the robustness of our approach even against adversarially hardened targets.

D. Analysis

1) Computational Efficiency: As shown in Fig. 4, we evaluate MF-CLIP's efficiency in terms of memory and time costs under two batch size settings. At batch size 1, MF-CLIP demonstrates minimal memory requirements (1,221 MB) and low time cost (3,449 s), significantly outperforming methods like DeCowA and L2T that consume up to 8,736 MB memory and 310,588 seconds. At batch size 128, MF-CLIP maintains efficient scaling with moderate memory usage (8,449 MB) and computation time (about 65 seconds), surpassing most baselines. Notably, in single-inference scenarios (batch size=1), MF-CLIP achieves near-optimal performance in both time and memory efficiency. When operating at batch size 128, while MF-CLIP's memory consumption is slightly higher than BIA, its superior attack performance more than compensates for this marginal overhead. This favorable efficiency profile stems from MF-CLIP's streamlined architecture, making it particularly suitable for scalable, resource-efficient applications.

2) Perceptual Quality Assessment: An important consideration in adversarial attack evaluation is the perceptual quality of generated perturbations, as imperceptibility remains a fundamental requirement for practical adversarial examples. We conduct a systematic quantitative analysis using two complementary perceptual metrics: the Structural Similarity Index Measure (SSIM) [46] and the Learned Perceptual Image Patch Similarity (LPIPS) [47]. SSIM quantifies structural similarity between original and perturbed images based on luminance, contrast, and structural information, while LPIPS leverages

TABLE III

COMPARISON BETWEEN MF-CLIP AND BASELINE METHODS WHERE THE SURROGATE MODELS ARE REPLACED WITH FINE-TUNED CLIP (WHEN THE BASELINE METHOD SUPPORTS SUCH MODIFICATION). DESPITE THESE ENHANCED BASELINES BENEFITING FROM BETTER SURROGATE MODELS, OUR METHOD STILL CONSISTENTLY ACHIEVES SUPERIOR PERFORMANCE ACROSS ALL DATASETS AND TARGET MODELS

	Method	Flowers102	Cars	Pets	Food101	SUN397	EuroSAT	UCF101
EfficientNet	SIA (ICCV 2023)	10.08	15.41	15.62	42.87	25.08	52.19	9.81
	DeCowA (AAAI 2024)	13.80	18.13	17.25	47.88	25.48	62.32	11.87
	BSR (CVPR 2024)	10.11	15.88	15.34	40.74	24.20	56.11	9.81
	L2T (CVPR 2024)	11.75	14.81	16.03	46.66	26.30	56.45	9.01
	MF-CLIP (ours)	**46.89**	**66.68**	**30.50**	**70.05**	**40.99**	**91.83**	**28.18**
RegNetX	SIA (ICCV 2023)	8.69	15.67	15.45	42.62	24.62	63.79	7.75
	DeCowA (AAAI 2024)	13.28	17.35	19.73	49.28	26.76	63.30	11.71
	BSR (CVPR 2024)	8.73	15.53	15.24	40.59	24.05	64.64	7.80
	L2T (CVPR 2024)	11.05	16.93	14.45	44.85	26.37	65.52	7.29
	MF-CLIP (ours)	**61.39**	**75.60**	**43.91**	**74.67**	**38.89**	**92.86**	**28.71**
ResNet-18	SIA (ICCV 2023)	5.03	7.65	11.28	26.80	15.23	56.62	3.57
	DeCowA (AAAI 2024)	8.36	10.88	14.69	34.89	17.28	63.91	7.30
	BSR (CVPR 2024)	4.95	8.31	11.94	24.89	14.53	58.74	4.28
	L2T (CVPR 2024)	5.72	8.10	13.45	30.42	16.79	60.30	3.98
	MF-CLIP (ours)	**56.07**	**55.69**	**41.67**	**70.14**	**38.44**	**91.77**	**27.68**

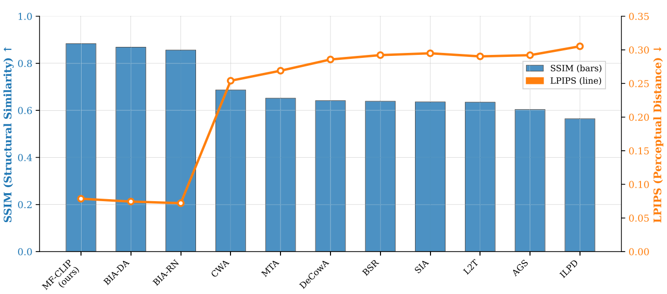


Fig. 5. Quantitative perceptual quality evaluation using SSIM and LPIPS metrics. SSIM values (bars) measure structural similarity preservation (higher values indicate better quality), while LPIPS values (line) assess perceptual distance (lower values indicate better quality).

deep neural network features to measure perceptual distance in a manner more aligned with human visual perception.

Fig. 5 reveals a clear performance stratification between different methodological paradigms. Generator-based approaches, including MF-CLIP and the BIA variants, consistently outperform optimization-based methods across both perceptual metrics. MF-CLIP achieves the highest SSIM score, demonstrating superior structural preservation, while the BIA variants excel in LPIPS evaluation. In contrast, gradient-based and input augmentation methods exhibit substantially lower perceptual quality across both metrics.

MF-CLIP demonstrates the optimal balance within the generator-based paradigm, achieving the highest structural similarity while maintaining competitive perceptual distance. This result, combined with our superior attack success rates, validates the practical effectiveness of margin-based fine-tuning for generating both effective and imperceptible adversarial examples.

E. Extended Evaluation

1) Enhanced Baseline Comparison: To further validate the effectiveness of our approach beyond the surrogate model selection, we conducted an additional experiment where we replaced the surrogate models in baseline methods with fine-tuned CLIP using supervised learning with cross-entropy loss (for those baseline methods that support modifying the surrogate model). This experiment aims to isolate the impact of our margin-based fine-tuning approach from the general benefits of using CLIP as a surrogate model.

As shown in Table III, even when baseline methods benefit from using fine-tuned CLIP as their surrogate model, MF-CLIP still consistently outperforms all competitors by substantial margins across all datasets and target models. Specifically, on the EfficientNet target model, MF-CLIP achieves improvements of 33.09% on Flowers102 compared to the best-performing enhanced baseline. Similar patterns are observed for RegNetX and ResNet-18 target models.

These results conclusively demonstrate that the superior performance of MF-CLIP stems not merely from the use of CLIP, but specifically from our margin-based fine-tuning approach that effectively enhances CLIP's discriminative capabilities for adversarial transferability. This finding supports our theoretical analysis that the margin in feature space plays a critical role in determining a model's effectiveness as a surrogate for adversarial attacks.

2) Large-Scale and Cross-Architecture Evaluation: To demonstrate the effectiveness in larger datasets and newer architectures like Vision Transformers, we conducted additional experiments on the ImageNet-1K validation set. As shown in Table IV, we evaluated MF-CLIP against baseline methods on four target architectures: EfficientNet-B0, RegNetX-1.6GF, ResNet-18, and importantly, ViT-B/16.

The results demonstrate that MF-CLIP substantially outperforms all baseline methods across all architectures. On ImageNet, our method achieves attack success rates of 59.96%, 59.40%, and 57.07% against EfficientNet, RegNetX, and ResNet-18 respectively. This confirms that our approach maintains its effectiveness on large-scale, diverse datasets.

TABLE IV

NO-BOX ATTACK PERFORMANCE ON IMAGENET VALIDATION SET ACROSS DIFFERENT ARCHITECTURES INCLUDING VISION TRANSFORMERS. ALL METHODS USE RESNET-50 BACKBONE AS SURROGATE MODEL. RESULTS SHOW THAT MF-CLIP CONSISTENTLY OUTPERFORMS BASELINES BY LARGE MARGINS, PARTICULARLY AGAINST VISION TRANSFORMERS

Method	EfficientNet	RegNet	ResNet-18	ViT-B/16
SIA	22.59	25.33	24.41	6.85
DeCowA	26.22	27.86	26.90	9.80
BSR	24.01	27.75	25.67	6.86
L2T	21.60	23.10	22.60	6.10
MF-CLIP (ours)	**59.96**	**59.40**	**57.07**	**24.92**

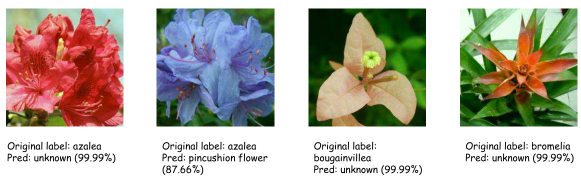


Fig. 6. Demonstration of MF-CLIP's effectiveness against industry-grade systems. Adversarial examples generated by our method successfully fool flower classification models on Baidu's PaddlePaddle platform.

These findings conclusively demonstrate that MF-CLIP's effectiveness extends beyond the limited datasets and architectures evaluated in prior work, establishing its utility across both large-scale natural image datasets and modern neural network architectures.

3) Real-World Case Study: To validate practical applicability, we tested MF-CLIP's adversarial examples against models deployed on Baidu's PaddlePaddle platform, a widely-used commercial framework in production environments such as license plate recognition and autonomous driving systems.

As shown in Fig. 6, we generated adversarial examples using MF-CLIP and tested them against flower classification models trained on the platform. All four adversarial examples successfully induced misclassification with high confidence scores (exceeding 87% in most cases), while maintaining visual imperceptibility to human observers.

This validation on commercial platforms demonstrates the practical security implications beyond academic benchmarks, confirming that MF-CLIP's approach exploits fundamental vulnerabilities in deep neural networks rather than architecture-specific weaknesses. The results highlight the importance of developing robust defenses against no-box attacks in production systems.

F. Ablation Studies

1) Surrogate Model Selection: We conduct comprehensive analytical experiments to evaluate vanilla CLIP's suitability as a surrogate model in the no-box setting. Fig. 7 presents a critical analysis about comparing different surrogate models with ResNet-50 backbone across multiple target datasets and architectures. The results reveal a surprising and counterintuitive phenomenon: despite CLIP's extensive pre-training on diverse data, vanilla CLIP not only fails to outperform ImageNet models when used as a surrogate, but actually performs worse in generating transferable adversarial examples.

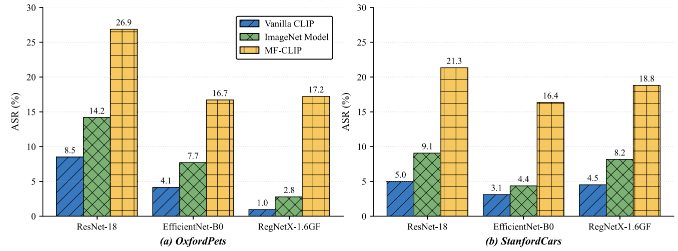


Fig. 7. Analysis for comparing different surrogate models with ResNet-50 backbone. The results clearly demonstrate that unrefined CLIP even performs worse than ImageNet models, while MF-CLIP significantly outperforms both.

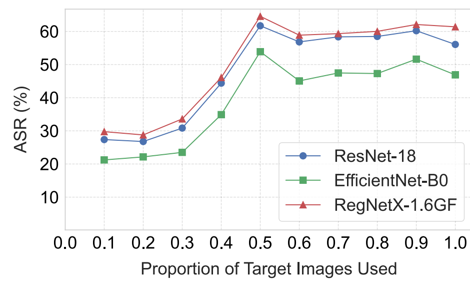


Fig. 8. ASR vs. proportion of target images used for fine-tuning on Flowers102 dataset.

These results provide strong empirical evidence supporting our theoretical analysis regarding CLIP's limited discriminative capabilities within specific domains. They highlight that while CLIP possesses superior representational capacity, this alone is insufficient for effective adversarial transferability. Most importantly, the dramatic performance improvement achieved by MF-CLIP over both vanilla CLIP and ImageNet models (as shown in Fig. 7) demonstrates the effectiveness of our margin-based fine-tuning approach in addressing this limitation, validating our method's core contribution.

2) Impact of Fine-Tuning Data Volume: To investigate MF-CLIP's sensitivity to the number of target images available for fine-tuning, we conduct experiments on the Flowers102 dataset using varying proportions (10% to 100%) of the test set.

3) Margin Parameter Analysis: The margin parameter m in Eq. (7) governs the angular separation between class representations in the embedding space, critically influencing attack transferability. To characterize this dependency, we evaluate $m \in [0, 0.5]$ across three distinct target architectures.

Fig. 9 reveals a consistent non-monotonic relationship between margin size and attack success rate. While standard softmax training ($m = 0$) yields suboptimal performance, introducing explicit margin constraints substantially enhances transferability, with optimal performance occurring within $m \in [0.05, 0.2]$. However, the performance drop at $m > 0.2$ indicates that excessive margin enforcement harms feature learning, reflecting the trade-off between discriminative and representational power [48]. When margins become overly restrictive, the model's capacity to capture intra-class variations essential for robust feature extraction is compromised.

The consistent optimal range across diverse architectures validates the stability of our margin-based approach while providing practical guidelines for hyperparameter selection.

TABLE V
COMPREHENSIVE ABLATION STUDY RESULTS FOR GENERATOR ARCHITECTURE VARIANTS, ORDERED BY PARAMETER COUNT. EXPERIMENTS CONDUCTED ON FLOWERS102 DATASET WITH ATTACK SUCCESS RATE AVERAGED ACROSS ALL TARGET MODELS

Variant	Encoder Channels (Input→Bottleneck)	Decoder Channels (Bottleneck→Output)	Processing Blocks	Attention Mechanism	Skip Conn.	Parameters (M)	FLOPs (G)	ASR (%)
V2	3→16→32→64	64→32→16→3	3	Multi-Head	✓	0.45	2.31	54.50
V1	3→32→64→128	128→64→32→3	1	Multi-Head	✓	1.06	6.50	54.60
V3	3→32→64→128	128→64→32→3	3	None	✓	1.59	8.07	54.68
V4	3→32→64→128	128→64→32→3	3	Multi-Head	✗	1.77	8.72	54.61
V5	**3→32→64→128**	**128→64→32→3**	**3**	**Multi-Head**	✓	**1.77**	**8.72**	**54.78**
V7	3→32→64→128	128→64→32→3	6	Multi-Head	✓	2.83	12.05	54.75
V6	3→64→128→256	256→128→64→3	3	Multi-Head	✓	7.04	33.84	54.92

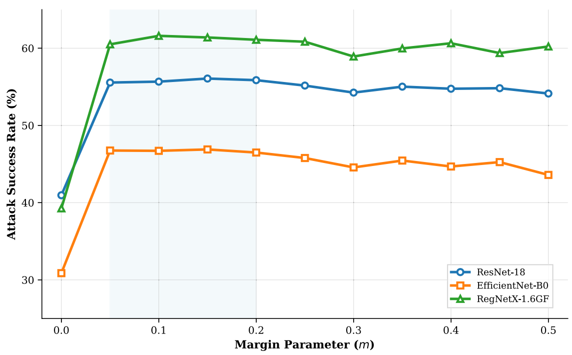


Fig. 9. Sensitivity analysis of the margin parameter m across three target architectures on Flowers102 dataset. The shaded region highlights the optimal margin range ($m \in [0.05, 0.2]$), with performance degrading for both smaller and larger values.

These findings confirm that moderate margin constraints effectively enhance inter-class separability without compromising the model's ability to learn generalizable representations.

4) Generator Architecture Design: To validate the architectural design choices of our generator network, we conduct systematic ablation studies examining the impact of different network configurations on adversarial transferability. We evaluate seven architectural variants by modifying individual components.

The experimental results in Table V reveal a remarkable finding: our method demonstrates robust insensitivity to generator architectural choices. Despite significant variations in model complexity, ranging from 0.45M to 7.04M parameters (15× difference) and 2.31G to 33.84G FLOPs (14× difference)—the attack success rates remain remarkably stable spanning only 0.42% from 54.50% to 54.92%.

This architectural robustness suggests that the effectiveness of MF-CLIP primarily stems from the margin-based fine-tuning of the surrogate model rather than sophisticated generator design. The consistent performance across diverse configurations indicates that once CLIP's discriminative capabilities are enhanced through our fine-tuning approach, relatively simple generator architectures suffice to produce effective adversarial perturbations. These findings have important practical implications, demonstrating that practitioners can select generator configurations based on computational constraints rather than performance considerations, thereby enabling flexible deployment across diverse resource environments while maintaining attack effectiveness.

V. CONCLUSION AND DISCUSSION

We presented MF-CLIP, a margin-based fine-tuning approach that enhances CLIP's effectiveness as a surrogate model for no-box adversarial attacks. Our analysis revealed CLIP's limitations in discriminative capabilities, which our method directly addresses. Experiments across diverse datasets, target models, and real-world settings demonstrate that MF-CLIP consistently outperforms state-of-the-art techniques, particularly at lower perturbation budgets.

Our findings reveal that improving a surrogate model's discriminative power is more effective than focusing solely on attack algorithms. MF-CLIP achieves remarkable cross-dataset generalization, maintaining effectiveness against adversarially trained models, Vision Transformers, and commercial API systems. These results provide insights into the relationship between representational capacity, discriminative power, and transferability in neural networks.

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